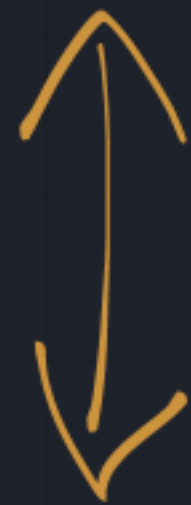


baalkikhaal.github.io.

physics

Engineering Physics



2025 Question paper.

CFET

$$E_{1e} = \frac{3}{2} k_B T$$

$$E_{NAe} = \frac{3}{2} k_B N_A T$$
$$= \frac{3}{2} R T$$

$$\therefore C_{\text{classical}} = \frac{d}{dT} \left(\frac{3}{2} R T \right)$$
$$= \frac{3}{2} R$$

$$C = \frac{dE}{dT}$$

QFET

(i) From QFET, only the fraction of electrons close to E_F only participate in thermalization. and the fraction is given by $\frac{k_B T}{E_F}$.

(ii) Increase in energy for one thermal electron = $k_B T$



From (i) & (ii), increase in energy per one electron of metal at temp T is $\Delta E = k_B T \cdot \frac{k_B T}{E_F}$ ★

Increase in energy per N_A electrons
of metal $\Delta E = k_B T \cdot \frac{k_B T}{E_F} \cdot N_A = \frac{k_B R T^2}{E_F}$

$$\therefore C_{\text{quantum}} = \frac{d'E}{dT} = \frac{\Delta E_{N_A}}{\Delta T}$$

$$= \frac{d}{dT} \left(\frac{k_B R}{E_F} T^2 \right)$$

$$= \frac{2 k_B R}{E_F} T$$

$$= 2 \left(\frac{k_B T}{E_F} \right) R$$

$$\approx C_{\text{classical}} \left(\frac{k_B T}{E_F} \right)$$

(c) $E_F = 5.5 \text{ eV}$, $T = 1000 \text{ K}$ $\longrightarrow$ If $k_B T_{300 \text{ K}} \rightarrow 26 \text{ meV}$
 $E = E_F + 0.3 \text{ eV} \Rightarrow E - E_F = 0.3 \text{ eV}$ then $k_B T_{1000 \text{ K}} \rightarrow ?$

$$P_{\text{occupy}}(E) = ?$$

$$\frac{26 \text{ meV} \cdot k_B T_{1000 \text{ K}}}{k_B T_{300 \text{ K}}} = \frac{26 \cdot 1000}{300} \text{ meV} = 86 \text{ meV}$$

Solⁿ: $P_{\text{occupy}}(E) = P_{\text{FD}}(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$
 $\approx \frac{1}{1 + \exp\left(\frac{0.3 \text{ eV}}{86 \text{ meV}}\right)} = \frac{1}{1 + \exp\left(\frac{300}{86}\right)} = 0.03$

⑥ (a) Electrical conductivity of intrinsic S.C.

$$\sigma(i-S.C) \begin{cases} \sigma(\text{electron channel}) \\ + \\ \sigma(\text{hole channel}) \end{cases} = \sigma_e + \sigma_h = n e \mu_e + p e \mu_h = n_i e \mu_e + n_i e \mu_h = n_i e (\mu_e + \mu_h)$$

But

$$(a^x)^y = a^{xy} \quad n = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right), \quad p = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right)$$

$$(a^2)^{\frac{1}{2}} = a^{2 \cdot \frac{1}{2}} = a^1$$

$$(a^2)^3 = a^2 \cdot a^2 \cdot a^2 = a^{2+2+2} = a^6 = a^{2 \cdot 3}$$

$$\text{and } n = p = n_i$$

so that

$$np = n_i \cdot n_i = n_i^2$$

$$= N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right) \cdot N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right)$$

$$= N_C N_V \exp\left(-\frac{E_C - E_F + E_F - E_V}{k_B T}\right) = N_C N_V \exp\left(-\frac{E_C - E_V}{k_B T}\right)$$

$$\Rightarrow n_i^2 = N_C N_V \exp\left(-\frac{E_g}{k_B T}\right)$$

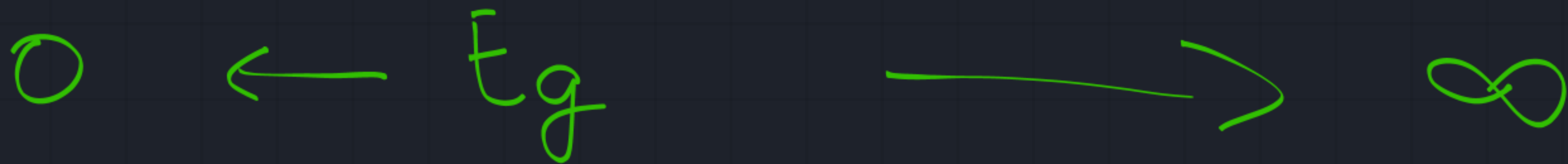
$$\Rightarrow n_i = \sqrt{N_C N_V} \exp\left(-\frac{E_g}{2 k_B T}\right)$$



$$\therefore \sigma_i = n_i e (\mu_e + \mu_h)$$

$$= \underbrace{\sqrt{N_c N_v} e (\mu_e + \mu_h)}_{\sigma_0} \exp\left(-\frac{E_g}{2k_B T}\right)$$

$$\sigma_i = \sigma_0 \exp\left(-\frac{E_g}{2k_B T}\right)$$



$$\sigma_i = \sigma_0$$

$$\therefore E_g(\text{Ge}) < E_g(\text{Si}) \Rightarrow n_i(\text{Ge}) > n_i(\text{Si})$$

$$\sigma_i = 0$$

$$\begin{matrix} \infty & -\infty \\ \text{"} & -1/8 \\ \text{"} & -1/8 \\ \text{"} & 0 \end{matrix}$$

$$\rho_i = 2.12 \Omega m \Rightarrow \sigma_i = \frac{1}{\rho_i} =$$

$$\text{Let } 2 \left(\frac{2\pi m_0 K_B T_{300}}{h^2} \right)^{3/2} = \cancel{Q}$$

$$\sigma_0 = \sqrt{N_c N_v} e (\mu_e + \mu_h)$$

$$\left. \begin{array}{l} N_c \propto m^*{}^{3/2} \\ N_c \propto T^{3/2} \end{array} \right\} \text{ then } N_c @ m_e^* = 0.5m_0, T = 600K$$

$$\mu_e = 0.36 \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$$

$$\mu_h = 0.17 \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$$

$$N_c = 2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}$$

$$N_v = 2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}$$

$$T = 300K$$

$$m_e^* = 0.5m_0, m_h^* = 0.37m_0$$

$$\propto \left(\frac{m^*}{m_0} \right)^{3/2} / \left(\frac{600}{300} \right)^{3/2}$$

$$(d) \quad \sigma(20^\circ\text{C}) = 2 \, \Omega^{-1}\text{m}^{-1}$$

$$E_g = 0.7 \text{ eV}$$

$$\sigma(40^\circ\text{C}) = ?$$

$$\text{Sol}^n \quad \sigma_i \propto \exp\left(-\frac{E_g}{2k_B T}\right)$$

$$\frac{\sigma(40^\circ\text{C})}{\sigma(20^\circ\text{C})} = \frac{\sigma(313\text{K})}{\sigma(293\text{K})} =$$

$$\frac{\exp\left(-\frac{E_g}{2k_B T_{313\text{K}}}\right)}{\exp\left(-\frac{E_g}{2k_B T_{293\text{K}}}\right)}$$

$$k_B T_{300\text{K}} = 26 \text{ meV}$$

$$k_B T_{313\text{K}} = ?$$

$$26 \text{ meV} \cdot \frac{313}{300}$$

$$k_B T_{293\text{K}} = 26 \text{ meV} \cdot \frac{293}{300}$$